

Math 6108 Homework 9

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Question 1.

Let $A \in \mathbb{C}^{n \times n}$ be a normal matrix. Then A is unitary if and only if every eigenvalue of A has absolute value 1.

Proof. Suppose that A is unitary. Let λ be an eigenvalue of A with corresponding eigenvector $\mathbf{v}$. Then, by the unitarity of A ,

$$1 = \mathbf{v}^* \mathbf{v} = \mathbf{v}^* A^* A \mathbf{v} = \mathbf{v}^* \bar{\lambda} \lambda \mathbf{v} = |\lambda|^2,$$

which implies that $|\lambda| = 1$.

Conversely, suppose that every eigenvalue of A has absolute value 1. Since A is normal, there exists a unitary matrix U such that

$$A = UDU^*, \quad D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix},$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . Then

$$D^* D = DD^* = \begin{bmatrix} |\lambda_1|^2 & & \\ & \ddots & \\ & & |\lambda_n|^2 \end{bmatrix} = I,$$

which implies that

$$\begin{aligned} A^* A &= (UDU^*)^* UDU^* = UD^* DU^* = UU^* = I, \\ AA^* &= UDU^* (UDU^*)^* = UDD^* U^* = UU^* = I. \end{aligned}$$

Thus, A is unitary. □

Question 2.

Let $A \in \mathbb{C}^{n \times n}$ be Hermitian, positive definite, and unitary. Then $A = I$.

Proof. Since A is Hermitian, it is normal; therefore, there exists a unitary matrix U such that

$$A = UDU^*, \quad D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix},$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . Since A is positive definite and Hermitian, its eigenvalues are all real and positive. On the other hand, since A is unitary, its eigenvalues all have absolute value 1 by Question 1. The only positive real number with absolute value 1 is 1 itself, so all the eigenvalues of A are 1, which means that $D = I$. Then

$$A = UDU^* = UU^* = I.$$

□

Question 3.

Let $A = \begin{bmatrix} 1 & x \\ x & 9 \end{bmatrix}$, where $x \in \mathbb{R}$. Then A is positive semidefinite if and only if $|x| \leq 3$.

Proof. The matrix A is positive semidefinite if and only if all of its eigenvalues are nonnegative. Any eigenvalue λ of A must be a root of the characteristic polynomial:

$$0 = \det(A - I\lambda) = (1 - \lambda)(9 - \lambda) - x^2 \iff \lambda^2 - 10\lambda + 9 - x^2 = 0.$$

By the quadratic formula, λ is an eigenvalue of A if and only if

$$\lambda = \frac{10 \pm \sqrt{100 - 4(9 - x^2)}}{2} = 5 \pm \sqrt{16 + x^2}.$$

Thus, A is positive semidefinite if and only if

$$5 \pm \sqrt{16 + x^2} \geq 0 \iff 5 - \sqrt{16 + x^2} \geq 0 \iff x^2 \leq 9 \iff |x| \leq 3.$$

□

Question 4.

Let $A \in \mathbb{C}^{m \times n}$, and let $P \in \mathbb{C}^{m \times m}$ and $Q \in \mathbb{C}^{n \times n}$ be unitary matrices. Then A and PAQ have the same singular values.

Proof. Let $A = U\Sigma V^*$ be a singular value decomposition of A . Then the diagonal elements of Σ are the singular values of A . On the other hand, $(PU)\Sigma(Q^*V)^*$ is a singular value decomposition of PAQ because

$$PAQ = PU\Sigma V^*Q = (PU)\Sigma(Q^*V)^*.$$

Then the singular values of PAQ are also the diagonal elements of Σ , meaning that A and PAQ have the same singular values. □

Question 5.

(a) Let

$$A = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

We first compute AA^* as

$$AA^* = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

The characteristic polynomial of AA^* is

$$\det(AA^* - I\lambda) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1).$$

Thus, the eigenvalues of AA^* are $\lambda_1 = 1$ and $\lambda_2 = 3$, and the singular values of A are 1 and $\sqrt{3}$. Let $\mathbf{u}_i$ be the eigenvector of AA^* corresponding to λ_i . Then

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \mathbf{u}_1 = \mathbf{u}_1, \quad \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \mathbf{u}_2 = 3\mathbf{u}_2$$

gives orthonormal eigenvectors $\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and $\mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Thus, the U and Σ matrices in a singular value decomposition $A = U\Sigma V^*$ of A can be chosen as

$$U = [\mathbf{u}_1 \quad \mathbf{u}_2] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \end{bmatrix}.$$

Since $A = U\Sigma V^*$, we find that

$$[A^*\mathbf{u}_1 \quad A^*\mathbf{u}_2] = A^*U = V\Sigma^T = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{bmatrix} = [\mathbf{v}_1 \quad \sqrt{3}\mathbf{v}_2],$$

so we can obtain the first two columns of V as

$$\mathbf{v}_1 = A^*\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{1}{\sqrt{3}} A^*\mathbf{u}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 & 1 \\ 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

The third column of V can be chosen freely as long as it is orthonormal with the first two columns, so

that V is unitary. We observe that $\mathbf{v}_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ works because

$$\langle \mathbf{v}_1, \mathbf{v}_3 \rangle = 0, \quad \langle \mathbf{v}_2, \mathbf{v}_3 \rangle = 0, \quad \langle \mathbf{v}_3, \mathbf{v}_3 \rangle = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} (1 + 1 + 1) = 1.$$

Thus, a singular value decomposition of A is given by

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \end{bmatrix} \begin{bmatrix} 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}^*.$$

(b) Let

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

We first compute

$$AA^* = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

The characteristic polynomial of AA^* is

$$\det(AA^* - I\lambda) = (3 - \lambda)((2 - \lambda)^2 - 1) = (3 - \lambda)(\lambda - 3)(\lambda - 1),$$

so the eigenvalues of AA^* are $\lambda_1 = 1$, $\lambda_2 = 3$, and $\lambda_3 = 3$. Then the singular values of A are 1, $\sqrt{3}$, $\sqrt{3}$. Let $\mathbf{u}_i$ be the eigenvector of AA^* corresponding to λ_i . Consider

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Then $3x = x$, so $x = 0$, and $2y + z = y$, so $z = -y$, and we can choose $\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. Now consider

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 3 \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Then x may be chosen freely, and $2y + z = 3y$ implies that $z = y$. Hence, we may choose $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$,

and $\mathbf{u}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Then $\mathbf{u}_1, \mathbf{u}_2$, and $\mathbf{u}_3$ are a set of orthonormal eigenvectors of AA^* , and the U matrix and Σ matrix in a singular value decomposition $A = U\Sigma V^*$ of A can be chosen as

$$U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3] = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{3} \end{bmatrix}.$$

As before, we have

$$[A^*\mathbf{u}_1 \quad A^*\mathbf{u}_2 \quad A^*\mathbf{u}_3] = A^*U = V\Sigma^T = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{3} \end{bmatrix} = [\mathbf{v}_1 \quad \sqrt{3}\mathbf{v}_2 \quad \sqrt{3}\mathbf{v}_3],$$

from which we may compute $\mathbf{v}_i$ for $i = 1, 2, 3$:

$$\begin{aligned} \mathbf{v}_1 &= A^*\mathbf{u}_1 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \\ \mathbf{v}_2 &= \frac{1}{\sqrt{3}} A^*\mathbf{u}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \\ \mathbf{v}_3 &= \frac{1}{\sqrt{3}} A^*\mathbf{u}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}. \end{aligned}$$

Then a singular value decomposition of A is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{3} \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \end{bmatrix}^*.$$